

Play and Learn – Project Proposal

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I will build Play and Learn alone in 4 weeks (Oct 1 – Oct 28, 2026): a website where students learn Python 101 and Java 101 through games and quizzes, and parents and admins can follow their progress.

Overview and goals

Play and Learn is an interactive course and quiz website. Students read short lessons, play coding games, and take quizzes. The site saves every score and gives each student feedback on what to practice next.

My goals for the project:

1. Help students learn coding concepts with games and interactive activities.
2. Test what students learned with quizzes that give instant feedback.
3. Track each student's progress in every lesson and course.
4. Give useful feedback based on each student's scores.
5. Show the system working with two reference courses: **Python 101 (Version 2.0)** and **Java 101 (Version 2.0)**.

I am working on this project alone, so I am responsible for all parts: the frontend, the backend, the database, and both courses.

Users and features

The website has three kinds of users. All three use the same login page, and each one only sees the pages for their role.

User	What they can do
Student	See a dashboard with their courses, progress and points. Read lessons, play a game for each lesson, and take a quiz. See right away if each answer is right or wrong, with an explanation. See charts of their scores and a list of topics to practice.

User	What they can do
Parent	See only their own child's progress: lessons finished, quiz and game scores, charts, and feedback.
Admin	Add, edit and delete courses, lessons, games, quiz questions and users. Link a student to a parent. See reports on every student and the questions students miss most.

The site has three kinds of games:

- **Matching:** click a line of code, then click what it prints.
- **Fill in the blank:** type the missing part of the code.
- **Put in order:** drag lines of code into the right order to build a working program.

Feedback is based on the student's scores. It names their average quiz score, shows how much they improved, lists the topics they missed (with a link back to that lesson), reminds them about quizzes below 70%, and points to the next lesson.

Course content

Each course has 8 lessons. Every lesson has 1 game and a 5-question quiz, so the site has 16 lessons, 16 games and 80 quiz questions in total. Students need 70% to pass a quiz.

#	Python 101 (v2.0)	Java 101 (v2.0)
1	Hello, Python! (print, comments)	Hello, Java! (class, main, println)
2	Variables and Data Types	Variables and Data Types
3	Operators and Math	Operators and Math
4	Working with Strings	Working with Strings
5	Making Decisions (if / elif / else)	Making Decisions (if / else / switch)
6	Loops (for and while)	Loops (for, while, do-while)
7	Lists	Arrays
8	Functions	Methods

Technology and database

Part	Tools
Frontend	HTML, CSS, JavaScript, jQuery, Bootstrap 5, Chart.js
Backend	PHP 8
Database	MySQL
Local server	XAMPP (Apache + PHP + MySQL)
Version control	GitHub

The database has 11 tables: `users` (role: admin, student or parent), `courses`, `lessons`, `games`, `game_items`, `quizzes`, `questions`, `lesson_progress`, `game_scores`, `quiz_attempts` and `attempt_answers`. A student is linked to a parent through `parent_id` in the `users` table. Each quiz question has a topic, which the site uses to tell students what to practice.

Security: passwords are stored with `password_hash()`, all database queries use prepared statements, all output is escaped, every form has a CSRF token, and quiz answers are checked on the server.

4-week schedule

I will show a working part of the project to the professor at the end of every week.

Week	Dates	Work	Demo to the professor
1	Oct 1 – Oct 7	Set up XAMPP and GitHub. Build the database (11 tables) and sample data. Build one login page for all 3 roles.	Log in as a student, a parent and an admin. Show the database tables.

Week	Dates	Work	Demo to the professor
2	Oct 8 – Oct 14	Build the Admin pages: add, edit and delete courses, lessons, games, quiz questions and users. Link students to parents. Load the Python 101 and Java 101 content.	Admin adds a new lesson and a quiz question. Both courses are loaded.
3	Oct 15 – Oct 21	Build the Student pages: dashboard, lessons, the 3 games, and quizzes with instant feedback. Save every score.	A student reads a lesson, plays a game and takes a quiz.
4	Oct 22 – Oct 28	Build the progress page with charts and feedback, and the Parent page. Test on phone and computer, fix bugs, and add security checks. Write the instructions and slides.	The full project from start to finish.

I plan to work about 1 to 2 hours every day so the work does not pile up at the end of the week.

Final deliverables

- The working website, with login for Student, Parent and Admin
- Python 101 and Java 101 fully loaded: 16 lessons, 16 games, 80 quiz questions
- The MySQL database files (`schema.sql` and `seed.sql`) with sample data
- A setup and user guide (README)
- Presentation slides and a live demo
- 4 weekly progress updates

Risks and how I will handle them

Risk	My plan
I am one person doing the work of three	I keep the scope to what is listed here. I work a little every day and finish each week's part before starting the next.
A bug blocks me for a long time	I save working versions on GitHub, so I can go back. I ask the professor or classmates for help early.
Something breaks during the demo	I practice the demo 2 to 3 times. The site works without internet, and a reset page reloads the sample data in one click.
I run out of time in Week 4	Charts and extra polish are the last things I add. The core parts (login, admin, lessons, games, quizzes, scores) are done first.